Sharvari Deshpande

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Education

Purdue University, John Martinsons Honors College — West Lafayette, IN
Bachelor of Engineering (BE), Computer Engineering — *Aug. 2023 – May 2027*

- 3.5 GPA, Dean's List, Semester Honors
- Certificate in Entrepreneurship and Innovation
- Undergraduate Teaching Assistant for ENGR 132: MATLAB Fundamentals (Spring 2025)
- Relevant Coursework: Data Structures and Algorithms, Python for Data Science, Electrical Engineering Fundamentals I & II, Digital Systems and Design, Signals and Systems, Linear Algebra, Differential Equations, Discrete Math

Experience

Engineering Intern — Jun. 2025 - Aug. 2025
Hawaiian Electric Company, Inc. — *Hilo, HI*

- Applied machine learning methods such as Ward's linkage clustering and Dynamic Time Warping in Python, SQL, and Apache Spark (Databricks) to raw voltage data, uncovering overvoltage trends to enhance monitoring.
- Led RTU replacement efforts at Kawailani Electrical Substation by performing circuit analysis and editing CAD wiring diagrams.
- Collaborated with engineers and contractors to monitor progress and manage project timelines.

NASA S.U.I.T.S. Team Jarvis Member — Aug. 2023 – Present
Space and Earth Analogs Research Chapter of Purdue (Purdue SEARCH) — *West Lafayette, IN*

- Co-authored a proposal for an autonomous rover system and AR astronaut interface, contributing to a section on optimal pathfinding and camera- and LiDAR-based localization methods.
- Selected as Team "JARVIS" to design proposed interface and systems for the NASA S.U.I.T.S. competitions.
- Awarded the "Pay It Forward" Award as Team JARVIS for impactful team contributions and STEM outreach, and received a NASA Artemis I mission flag that orbited the Moon aboard the Artemis I mission rocket.

Undergraduate Data Science Researcher, Viasat Inc. — Aug. 2024 – Jan. 2025
The Data Mine - Purdue University — *West Lafayette, IN*

- Co-developed and automated a data pipeline using Python and R to collect and clean satellite data from DiscosWeb API for advanced geospatial analysis.
- Co-presented critical satellite data findings to Viasat Inc. company representatives.

Undergraduate Data Analyst — Oct. 2023 – May 2025
Purdue University — *West Lafayette, IN*

- Performed data ETL operations in Excel to ensure accuracy of Purdue University's student admissions datasets for submission to college ranking platforms (e.g. U.S. News.)
- Utilized advanced Excel functions, such as PivotTables and formulas, to extract insights.
- Collaborated with Purdue University's Director of Data Analytics and Information to review findings.

Projects

NASA S.U.I.T.S.: AR Headset for EVAs: Interactive Tasklist — Jan. 2024 – May 2024
Space and Earth Analogs Research Chapter of Purdue (Purdue SEARCH) — *West Lafayette, IN*

- Utilized C# and Python on platform UNITY to develop an interactive task-list responsible for managing and simulating an astronaut's command system during the Artemis missions.

Audio Equalizer — Apr. 2025 – May 2025
ECE 20007 Final Project — *West Lafayette, IN*

- Designed and built a circuit-based analog multi-band audio equalizer with adjustable bass, mid-range, and treble using RC low-, band-, and high-pass filters and op-amps.
- Simulated and validated complete circuit via SPICE tools prior to hardware implementation.
- Characterized frequency response and output power using oscilloscope-based measurements.

Technical Skills

Languages: Python (Pandas, NumPy, Matplotlib), Java, SQL, R, C, C++, C#, HTML, LaTeX
HDL & Digital Design: Verilog RTL, FPGA-based Design, SPICE (LTspice, PSpice), Finite State Machines (FSMs), Combinational & Sequential logic, RTL Datapaths and Control, Timing Analysis & Verification, KiCAD
Developer Tools: Git, Databricks, PySpark, Apache Spark, UNITY, VS Code, Visual Studio, Linux, Eclipse, Microsoft Excel (Advanced Functions), Microsoft Office Suite, MATLAB, JupyterNotebook, Kanban